

Date: 2026-07-29
Request ID: 100048800

Microwave				
Analyte(s)	Mn			
SOP#				
Acid Cocktail	6 mL HCl, 2 mL HNO3			
Rack	7			
Vessel	PTFE			
Stir	90%			
Time (min)	Power (W)	T1 (°C)	T2 (°C)	P (bar)
15	1200	180	65	110
15	1500	240	65	110
30	1500	240	65	110
sample(s)	weight (g)	volume (mL)		
200128666_1	0.1065	250		
200128666_2	0.1030	250		
200128667_1	0.1090	250		
200128667_2	0.1068	250		
200128668_1	0.1054	250		
200128668_2	0.1044	250		
200128669_1	0.1083	250		
200128669_2	0.1050	250		

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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128666	17.2377	19.2366	18.8745	18.11	0.8189

Mn ICP analysis

sample	Dilution	conc. [mg/L]	Mn_ppm [dried]	%Mn [dried]
200128666_1	100/0.84	1.857	633750.73	63.38
200128666_2	100/0.84	1.788	630761.24	63.08

mn result: 632255.98 ppm

mn result: 63.23 %

mn lot: *Manufacturer: Assurance, 28-137MNY exp: 2027-01-30*

Note(s):

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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128667	17.1555	19.1844	18.8096	18.47	0.8153

Mn ICP analysis

sample	Dilution	conc. [mg/L]	Mn_ppm [dried]	%Mn [dried]
200128667_1	100/0.84	1.918	642364.76	64.24
200128667_2	100/0.84	1.849	631841.00	63.18

mn result: 637102.88 ppm

mn result: 63.71 %

mn lot: *Manufacturer: Assurance, 28-137MNY exp: 2027-01-30*

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sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128668	15.5590	17.5580	17.2425	15.78	0.8422

Mn ICP analysis

sample	Dilution	conc. [mg/L]	Mn_ppm [dried]	%Mn [dried]
200128668_1	100/0.84	1.905	638810.07	63.88
200128668_2	100/0.84	1.875	634604.66	63.46

mn result: 636707.37 ppm

mn result: 63.67 %

mn lot: *Manufacturer: Assurance, 28-137MNY exp: 2027-01-30*

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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128669	18.7973	20.7939	20.4750	15.97	0.8403

Mn ICP analysis

sample	Dilution	conc. [mg/L]	Mn_ppm [dried]	%Mn [dried]
200128669_1	100/0.84	1.973	645343.80	64.53
200128669_2	100/0.84	1.913	645302.21	64.53

mn result: 645323.01 ppm

mn result: 64.53 %

mn lot: *Manufacturer: Assurance, 28-137MNY exp: 2027-01-30*

Note(s):

$$\text{ppm } M+ = [\text{conc.}][\text{volume}][\text{dilution}]/[\text{weight}]$$

$$\%M+ = [\text{conc.}][\text{volume}][\text{dilution}]/([\text{weight}]*10,000)$$

$$\%MO = [\text{oxide factor}]*\%M+$$

A = crucible

B = crucible + sample

C = crucible + sample after drying

$$\%LOI = ([B-A]-[C-A])/(B-A)*100\%$$

$$\text{ppm } M+ \text{ [dried]} = \text{ppm } M+ / ([1-(\%LOI)/100])$$

$$\%M+ \text{ [dried]} = \%M+ / ([1-(\%LOI)/100])$$

$$\%MO \text{ [dried]} = \%MO / ([1-(\%LOI)/100])$$